

Broad-band PSF errors inflate little red dot hosts

Source-conditioned polychromatic PSFs reveal systematic host overestimation in compact high-redshift nuclei.

arXiv:2608.30459; Zhang et al. LRDs; PSF modeling; high-redshift compact sources.

First author / year / source

arXiv:2608.30459 doi:10.48550/arXiv.2608.30459



Background

Compact high-redshift sources often overlap with distinct physical components in diffraction-limited images. Field stars are used to model the point-spread function and infer host structure.

Broad-band PSF modeling assumes field stars match target spectra, but spectral mismatches can misassign light.

What this paper is selling

The paper shows that broad-filter PSF fitting systematically overestimates extended host flux for spectra differing from field stars. It derives the error's origin and provides a correction for other compact distant sources.

Key results to remember

- Point-only LRDs produce wavelength-dependent residuals mimicking Sérsic hosts in broad filters.
- The F444W host allocation exceeds 11.8% for every tested spectral and detector variation.
- Spectral mismatch between target and PSF stars drives the systematic host overestimate.

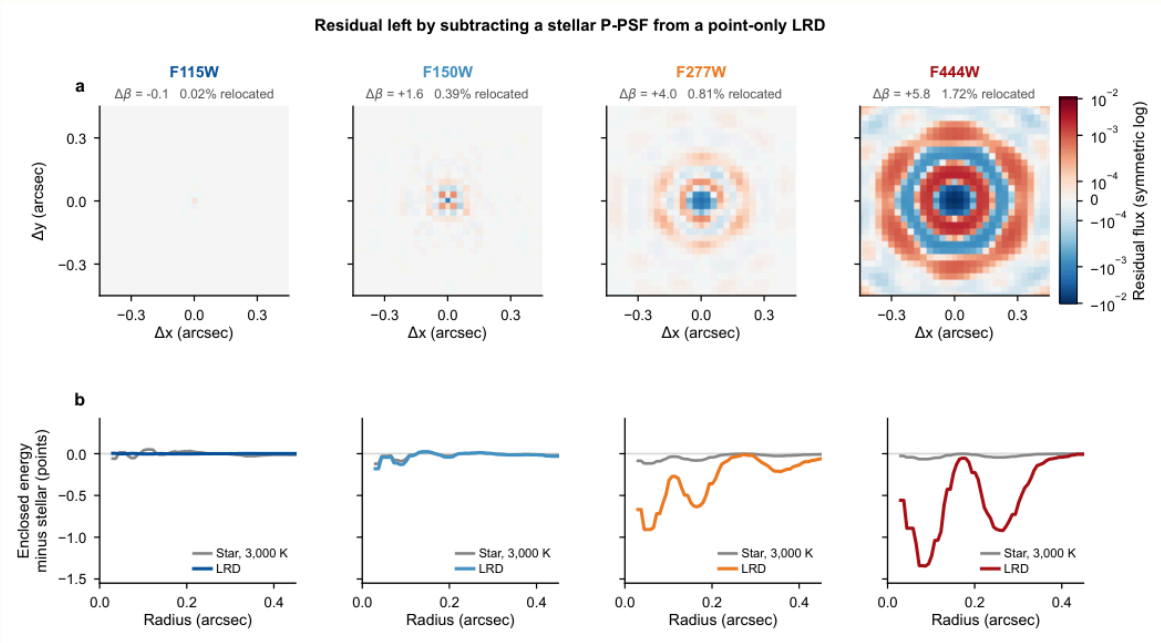


Fig. 1. A point-only LRD yields filter-dependent residuals that are wrongly attributed to a Sérsic host component.

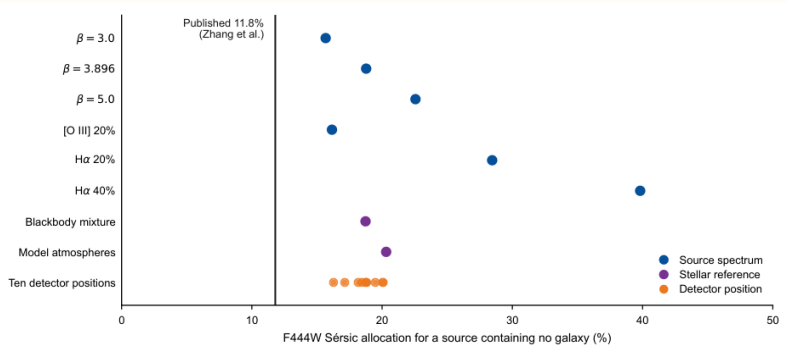


Fig. 2. All tested spectral and detector variations allocate more than 11.8% F444W flux to a host.